



Hyweb ECR API Integration Specification

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Revision History

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1.2	2018/01/18	Update the example on 2.4.2 Adding supporting on returning campaign id for coupon usage on Section 4.2,4.8,4.20	David
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		Updated Section 5.2,5.6,5.8,5.10,5.12,5.14,5.20 for return extra fields: AID, EntryMode, BatchNo Updated Section 5.1,5.5 for handling empty PAYMENT_APP_ID Adding Section 5.23,5.24 for new Reversal API Updated Section 6.3,6.4 for new Reversal API Updated Section 2.4.4, 3.4.4, 4.4.3 for exception handling	
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1.18	2019/12/26	Updated in Section 6.3 to support Wechat/Alipay refund processing status	Rex
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1.29	2021/03/18	Added Section 5.39 & 5.40	Rex
1.30	2021/04/07	Updated Section 5.1 to support UPLAN sale	Rex



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1. Introduction

1.1 Purpose

This document describes the interface between Electronic Cash Register (ECR) and the Hyweb N5 terminals to accept card transaction and Third Parties Payment transaction like Wechat Pay, Alipay and etc.

1.2 Scope

This document includes the below integration method

- Android App Call App Integration
- Serial port Integration
- TCP/IP Socket



2. Integration Method – Android App Call App

2.1 System Overview

ECR vendor will install an android application (ECR MobileApp) into the N5 Terminal (EDC) and integrated with the Payment Application by using Android App Call App Approach

2.2 Assumption

The ECR MobileApp will connect with ECR Server via Internal Wi-Fi and this internal Wi-Fi could connect to internet as well (for Payment Application to connect to Payment Host and Terminal Management System)

2.3 Android Requirement for ECR MobileApp

Android OS Version: 5.1

Android SDK Min Version: 19



2.4 Communication Interface

2.4.1 App Call App Protocol

ECR MobileApp and Terminal need to provide the *Android Package id* and *Android Activity Class Name* to exchange data in between. Those data will be stored inside Android Intent Class.

Intent Field Name	Description
COMMAND_TYPE	For Message from ECR to Terminal : ECR-REQ For Message from Terminal to ECR: ECR-RESP
DATA	In JSON String Format (Refer to Section 4)
MAC	Refer to 2.4.2 for Calculation

2.4.2 Error Handling

Operation timeout: EDC will return to IDLE stage and send Response Message with Status Code “98” to ECR when there is operation timeout. ECR should treat this transaction as failure.

EDC busy handling: When ECR sends request to EDC which is not in IDLE stage, ECR will receive response with Status Code “97”. ECR should prompt message to ask user to complete current transaction in EDC before send another request.

App crash handling: When ECR found there is some missing response from EDC (which may due to application crash). ECR should send the transaction retrieval request command to EDC to get the last transaction.



3. Integration Method – Serial Port Integration

3.1 System Overview

ECR will connect N5 Terminal via

- COM Port to RS232
- USB to RS232 (need to install USB-COM driver)

3.2 Communication Setting

This is a serial/ASYNC communications, by default the settings will be as below

Parameter	Value
Baud rate	9600
Parity	None
Data bit	8
Stop bit	1
Flow control	None

3.3 Message Frame

All messages will be framed using the following format

Field	Value	Description
STX	02H	Start of Text
Message Length	LLLL	2 Bytes Length Indicator, BCD
Message		In JSON String Format (Refer to Section 5)
ETX	03H	End of Text
LRC		<u>Longitudinal Redundancy Check</u> Checksum from every character in the transaction message after the STX and



		including the ETX, calculated by XOR every bytes.
--	--	---

3.4 Transmission Protocol

In addition to the above described messages, ACK (06H) and NAK (15H) control characters are also required to ensure error free exchange of request and response messages. An ACK indicates the successful reception of a message; a NAK indicates that the receiver requests the retransmission of the last message that was received in error.

The ACK and NAK characters are expected to be received within 2 seconds from the transaction of a message. Every message is expected to get an ACK or NAK. A message can be sent again (up to 3 times) after the 2 second ACK/NAK response time has expired. In theory, not all request messages generate a response message but they all require to be ACKed.

3.4.1 Good LRC

ECR	Direction	POS
Request Message	----->	
	<-----	ACK
	<-----	Response Message
ACK	----->	

3.4.2 Bad LRC

ECR	Direction	POS
Request Message	----->	
	<-----	NAK
Request Message	----->	
	<-----	ACK
	<-----	Response Message
NAK	----->	



	←-----	Response Message
ACK	-----→	

3.4.3 Timeout

ECR	Direction	POS
Request Message	-----→	
		No ACK/NAK after 2s
Request Message	-----→	
	←-----	ACK
	←-----	Response Message
No ACK/NAK after 2s		
	←-----	Response Message
ACK	-----→	

3.4.4 Error Handling

NAK handling, If ECR receive NAK from after sending request message, it should issue the original request message to EDC until receiving ACK from EDC

No ACK/NAK timeout: 2s, If ECR does not receive ACK/NAK from EDC in 2s after sending request message, it should issue the original request message to EDC until receiving ACK from EDC

Operation timeout: 120s, If ECR does not receive any response from EDC in 120s, it should issue the transaction retrieval request command to EDC to get the last transaction or the ECR should allow user to press button for transaction retrieval to prevent message lost. However if ECR receives response message with Status Code "98", ECR should treat this transaction as failure.

EDC busy handling: When ECR sends request to EDC which is not in IDLE stage, ECR will receive response with Status Code "97". ECR should prompt message to ask user to complete current transaction in EDC before send another request.



3.4.5 Link Test

ECR can use Link Test message to verify the connection status between ECR and EDC



4. Integration Method – TCP/IP Socket & Bluetooth

4.1 System Overview

ECR vendor need to develop a socket client model in POS program and integrate with the Payment Application in N5 Terminal by using TCP/IP Socket Approach

4.2 Communication Setting

This is a TCP/IP socket communication, by default the settings will be as below

Parameter	Value
IP	Dynamically configurable e.g. 192.168.1.99
Port	9001

This is a Bluetooth communication, by default the settings will be as below.

Parameter	Value
UUID	“00001101-0000-1000-8000-008”+[profile_id]
MAC Address	Bluetooth Mac Address of terminal

4.3 Assumption

The N5 terminal will be connected via Internal Wi-Fi either by ECR Server or the ECR POS program. ECR POS program will connect with ECR Server via Internal Wi-Fi and this internal Wi-Fi could connect to internet as well (for Payment Application to connect to Payment Host and Terminal Management System)

4.4 Interface

4.4.1 Message

Field	Value	Description
STX	02H	Start of Text



Message Length	LLLL	2 Bytes Length Indicator, BCD
Message		In JSON String Format (Refer to Section 5)
MAC		16 Bytes, Refer to Section 4.4.2
ETX	03H	End of Text

4.4.2 Error Handling

Socket timeout: 120s, If ECR does not receive any response from EDC in 120s, it should issue the transaction retrieval request command to EDC to get the last transaction or the ECR should allow user to press button for transaction retrieval to prevent message lost. However if ECR receives response message with Status Code “98”, ECR should treat this transaction as failure.

EDC busy handling: When ECR sends request to EDC which is not in IDLE stage, ECR will receive response with Status Code “97”. ECR should prompt message to ask user to complete current transaction in EDC before send another request.

4.5 Mac Calculation

To ensure the data authenticity of message between ECR & terminal. Both of ECR & terminal need to verify mac. There are three steps. Firstly, build source message according to the rule. Then use the source message to calculate Mac. Rebuild the request message by original Json & Mac.

4.5.1 Build Source Data

Build the Json message(Refer to Section 5) to request terminal. Then put the MacKey(Please request with Hyweb) in ECR application configuration.

For Example,

Sale with amount:37

```
{ "EVENT_NAME": "SALE", "TXN_ID": "id_000000000000000001", "TXN_AMT": 37, "TIPS": 0, "PAYMENT_APP_ID": "", "QRC_VALUE": "", "PAN": "", "EXPIRY_DATE": "", "CARD_TYPE": "", "PRINT_SS": false }
```

Combine Json message and MacKey with ‘&’

```
{ "EVENT_NAME": "SALE", "TXN_ID": "id_000000000000000001", "TXN_AMT": 37, "TIPS": 0, "PAYMENT_APP_ID": "", "QRC_VALUE": "", "PAN": "", "EXPIRY_DATE": "", "CARD_TYPE": "", "PRINT_SS": false } & [MACKEY]
```

MD5 (technically called MD5 Message-Digest Algorithm) is a cryptographic hash function whose main purpose is to verify that a message/file has been unaltered.

MD5 (technically called MD5 Message-Digest Algorithm) is a cryptographic hash function whose main purpose is to verify that a message/file has been unaltered.

For Example,

Json =

```
{
  "EVENT_NAME": "SALE",
  "TXN_ID": "id_000000000000000001",
  "TXN_AMT": 37,
  "TIPS": 0,
  "PAYMENT_APP_ID": "",
  "QRC_VALUE": "",
  "PAN": "",
  "EXPIRY_DATE": "",
  "CARD_TYPE": "",
  "PRINT_SS": false
}
```

```
Mac = "E9C553B8269F1DB701AAF870C264C134"
```

Step1. Build msg by starting with STX and message length

[illegible]

Step2. Add MAC

[illegible]

Step3. Add ETX

[illegible]



5. API Interfaces

This is the full API Interfaces which may not appropriate for your ECR, please discuss with our technical team to confirm the integration scope.

◆ Transaction Currency:

156 : RMB

344 : HKD

446 : MOP

840 : USD

116 : KHR

418 : LAK

Note : Transaction Currency is fixed per each terminal

◆ Time:

UTC+08:00

◆ TXN_ID:

Unique Transaction ID provided by ECR which is up to 32 characters

e.g. 1217752501201407033233368018

◆ Type to String Conversion

Type	Conversion Example
Decimal	If the value is 123.45, then the converted string will be 123.45
Boolean	If the value is true, the converted string will be Y If the value is false, the converted string will be



	<i>N</i>
--	----------



5.1 Sale Request

Field	Java Object	Max Length	Required	Description
EVENT_NAME	String	4	Y	SALE
TXN_ID	String	32	Y	Unique Txn ID generated by ECR
TXN_AMT	Decimal	12	Y	Transaction Amount(e.g. 100.00)
TIPS	Decimal	12	N	Applied if tip is enabled(e.g. 100.00)
PAYMENT_APP_ID	String	10	N	Refer to Payment Application List If empty, Teller will prompt dialog to choose paymentApp. For UPLAN, please use "CPN", terminal will send a CouponUse request to host first. Then terminal will open read card screen to perform a card sale for remained amount (txn amount – discount amount)
QRC_VALUE	String	150	N	For CC/QRC, it is the Customer Presented QR Code (CPQRC) Value For CPN 1. it is the Coupon QR/Barcode Value 2. it is the UPLAN QR/Barcode Value
MERCHANT_REF	String	10	N	ECR needs to send to N5 only If it needs to print this ref on n5 slip
PAN	String	19	N	For CC/Amex, used for manual input entrymode
EXPIRY_DATE	String	4	N	For CC/Amex, used for manual input entrymode YYMM
CARD_TYPE	String	3	N	For CC/Amex only. VC: Visa Card MC: MasterCard JCB: Japan Credit Bureau Card UP: Union Pay AE: Amex

5.2 Sale Response

Field	Java Object	Max Length	Required	Description
EVENT_NAME	String	9	Y	SALE_RESP
TXN_ID	String	32	Y	Unique Txn ID per each request defined by ECR
STATUS	String	2	Y	Refer to Status Code List



TXN_AMT	Decimal	12	Y	Transaction Amount (Original) However, it may not equal to Authorization Amount
TIPS	Decimal	12	N	Applied if tip is enabled
PAYMENT_TYPE	String	10	N	Refer to Payment Type List
RESP_CODE	String	32	N	Only exists if Status = 00/10 It is the authorization response code (or UPLAN/LMS resp code if failed), For Wechat/Alipay, it is the err_code If Status = 00 It is the error code of printing. Z0001 Printer error Z0002 Printer no paper
Below fields will be returned only if STATUS = 00				
LOYALTY_TYPE	String	2	N	If used Coupon/loyalty, this field will be returned UP - Uplan MT - Merchant Coupon BK - Bank Coupon LM - LMS
CAMPAIGN_ID	String	30	N	If used Coupon/loyalty, the related campaign id will be returned
TRACE_NO	String	32	Y	For CC, it is the trace no on sale slip (e.g. invoice no from TAS) For Wechat/Alipay, it is the 商戶交易單號
TXN_DATE	String	8	Y	Transaction Date in YYYYMMDD format
TXN_TIME	String	8	Y	Transaction Date in hhmmss format
HOST_REF	String	32	Y	For CC, it is RRN For Wechat/Alipay, it is the transaction id from 第三方
COUPON_ID	String	19	N	Applied if Loyalty Type is UP/MT/BK Coupon ID
AUTH_CODE	String	8	N	For CC only, authorization code
AUTH_AMT	Decimal	12	N	Txn_amt + tips - discount (refer below)
DISCOUNT	Decimal	12	N	Applied if Loyalty Type has value Discount Amount
LOCAL_CUR	String	3	N	Transaction Currency code e.g. 344
FOREIGN_CUR	String	3	N	Cardholder Currency code e.g. 156
FXRATE	Decimal		N	Exchange Rate
FOREIGN_AMT	Decimal		N	Foreign Transaction Amount



MID	String	32	Y	For CC: MID For Wechat/Alipay: sub_mch_id
TID	String	32	Y	For CC: TID For Wechat/Alipay: terminal_id
PAN	String	19	N	For CC only Masked PAN
EXPIRY_DATE	String	4	N	For CC only Masked Expiry Date YYMM
LOYALTY_REF	String	32	N	MT/BK -商戶訂單號 LM – LMS Reference (12 digits)
Below fields will be returned only if STATUS = 00 && ECR needs to print receipt				
AID	String	32	N	For CC & Amex only
BATCH_NO	String	6	N	For CC & Amex only
TC	String	30	N	For CC & Amex only
APP	String	30	N	For CC & Amex only
ACQUIRER	String	30	N	For CC & Amex only
IF_SIGN	String	1	N	For CC & Amex only
CARD_TAG	String	30	N	ForCUP cards only Credit: (C) Debit: (D) Sample_CUP_Sale(Credit): CARDNUMBER/TYPE 621234*****0201(T)(C)
CARD_TYPE	String	5	N	For CC & Amex only Chip: (C) RF: (T) SWIPE: (S) FALLBACK: (F) QR: (Q) MANUAL: (M)



				Sample_VMJ_Sale(Chip): CARDNUMBER/TYPE 621234*****0201(C) Sample_CUP_Sale(RF): CARDNUMBER/TYPE 621234*****0201(T)(C)
ENTRY_MODE	String	5	N	For CC & Amex only Chip: (I) RF: (C) SWIPE: (S) FALLBACK: (F) QR: (Q) MANUAL: (M)
SIGN_DESC	String	50	N	For CC If need customer sign, return "持卡人簽名 CARDHOLDER SIGNATURE". If no need customer confirm, return "無須簽署 NO SIGNATURE REQUIRED". For Amex If need customer sign, return "持卡人簽名 CARDHOLDER SIGNATURE". If no need customer confirm, return "無須簽署 NO SIGNATURE REQUIRED". If use AEQR, return "已通過持卡人驗證 Verified by CDCVM" If in input pin, Return "已驗證密碼 PIN VERIFIED"
TVR	String	10	N	For CC & Amex only, only shows when insert or tap card
TSI	String	4	N	For CC & Amex only, only shows when insert or tap card



CARD_HOLDER	String	100	N	For CC only, when APP = VISA/MasterCard/JCB and entry mode is not RF
TERMS	String	300	N	For CC & Amex only PLEASE USE 5.29 PARAMS RQUEST to get below params. #tc1: "本人確認以上交易，同意將其記入本 卡賬戶 I ACKNOWLEDGE SATISFACTORY RECEIPT OF RELATIVE GOODS/SERVICES. I AGREE TO PAY THE ABOVE TOTAL AMOUNT (TOGETHER WITH ANY OTHER CHARGES DUE THEREON) ACCORDING TO THE CARD ISSUER AGREEMENT."
DCC_VISA_TERMS	String	300	N	For DCC only, CUP does not support DCC PLEASE USE 5.29 PARAMS RQUEST to get below params. #dcc1: "匯率 Rate : " #dcc2: "Mark[X] Transaction Currency" #dcc3: "THIS RECEIPT COMPLIES WITH VISA RULES AND THE SERVICE IS PROVIDED BY SERVICE PROVIDER I'VE BEEN OFFERED CHOICE OF CURRENCIES INCLUDING" #dcc4: "PERCENT OVER WHOLE SALE RATE." VISA DCC "#dcc1 HKD/USD 0.133500 #dcc2 [] HKD Amount USD Amount[] HKD1500.00 USD200.25 #dcc3 HKD*INCL.THREE PT. FIVE ZERO #dcc4" MASTER/JCB DCC "#dcc1 HKD/USD 0.133500 #dcc2 [] HKD Amount USD Amount[] HKD1500.00 USD200.25
DCC_MID	String	32	N	DCC only Displayed under MID



REMAINED_VALUE	String	20	N	For OPS only
SERIAL_NO	String	10	N	For OPS only
LMS_ORIGINAL_BAL	String	20	N	原有 LMS 積分 For LOYALTY_TYPE=LM only
LMS_BAL	String	20	N	結餘 LMS 積分 For LOYALTY_TYPE=LM only
LMS_USED	String	20	N	使用 LMS 積分 For LOYALTY_TYPE=LM only
LMS_BATCH_NO	String	10	N	For LOYALTY_TYPE=LM only
LMS_APPRV	String	10	N	For LOYALTY_TYPE=LM only

5.3 Void Request

Field	Java Object	Max Length	Required	Description
EVENT_NAME	String	4	Y	VOID
TXN_ID	String	32	O	Original Transaction ID For CC/Amex, it is required.



				For QRC, either TXN_ID or QRC_VALUE must be sent.
QRC_VALUE	String		O	For QRC only. For Wechat/Alipay, it is the Transaction QR/Barcode (QR printed in sale slip or barcode in wallet's transaction history)
MERCHANT_REF	String	10	N	ECR needs to send to N5 only If it needs to print this ref on n5 slip

5.4 Void Response

Field	Java Object	Max Length	Required	Description
EVENT_NAME	String	9	Y	VOID_RESP
TXN_ID	String	32	Y	Original Transaction ID
STATUS	String	2	Y	Refer to Status Code List
PAYMENT_TYPE	String	10	N	Refer to Payment Type List
RESP_CODE	String	32	N	Only exists if Status = 00/10 For CC, it is the authorization response code (or UPLAN resp code if failed), For Wechat/Alipay, it is the err_code If Status = 00 It is the error code of printing. Z0001 Printer error Z0002 Printer no paper
Below fields will be returned only if STATUS = 00				
TXN_DATE	String	8	Y	Void Transaction Date in YYYYMMDD format
TXN_TIME	String	8	Y	Void Transaction Time in hhmmss format
HOST_REF	String	32	Y	For CC, it is RRN of Void Txn For Wechat/Alipay, it is the void transaction id from 第三方
Below fields will be returned only if STATUS = 00 && ECR needs to print receipt				
BATCH_NO	String	6	N	For CC & Amex only
MID	String	32	Y	For CC: MID For Wechat/Alipay: sub_mch_id
TID	String	32	Y	For CC: TID For Wechat/Alipay: terminal_id
PAN	String	19	N	For CC only Masked PAN



TXN_AMT	Decimal		N	
TRACE_NO	String	32	Y	For CC, it is the trace no on sale slip (e.g. invoice no from TAS) For Wechat/Alipay, it is the 商戶交易單號
AUTH_CODE	String	8	N	For CC only, authorization code
CARD_HOLDER	String	100	N	For CC only, when APP = VISA/MasterCard/JCB and when entry mode is not contactless
APP	String	30	N	For CC & Amex only
ACQUIRER	String	30	N	For CC & Amex only
TERMS	String	300	N	For CC & Amex only PLEASE USE 5.29 PARAMS RQUEST to get below params. #tc1:"本人確認以上交易，同意將其記入本卡賬戶 I ACKNOWLEDGE SATISFACTORY RECEIPT OF RELATIVE GOODS/SERVICES. I AGREE TO PAY THE ABOVE TOTAL AMOUNT (TOGETHER WITH ANY OTHER CHARGES DUE THEREON) ACCORDING TO THE CARD ISSUER AGREEMENT."
DCC_VISA_TERMS	String	300	N	For DCC only, CUP doest not support DCC PLEASE USE 5.29 PARAMS RQUEST to get below params. #dcc1:"匯率 Rate : " #dcc2:"Mark[X] Transaction Currency" #dcc3:"THIS RECEIPT COMPLIES WITH VISA RULES AND THE SERVICE IS PROVIDED BY SERVICE PROVIDER I'VE BEEN OFFEREDCHOICE OF CURRENCIES INCLUDING" #dcc4:"PERCENT OVER WHOLE SALE RATE." VISA DCC "#dcc1 HKD/USD 0.133500 #dcc2 []HKD Amount USD Amount[] HKD1500.00 USD200.25 #dcc3 HKD*INCL.THREE PT. FIVE ZERO #dcc4" MASTER/JCB DCC



				"#dcc1 HKD/USD 0.133500 #dcc2 []HKD Amount USD Amount[] HKD1500.00 USD200.25
DCC_MID	String	32	N	DCC only Displayed under MID 
LMS_ORIGINAL_BAL	String	20	N	原有 LMS 積分 For LOYALTY_TYPE=LM only
LMS_BAL	String	20	N	結餘 LMS 積分 For LOYALTY_TYPE=LM only
LMS_USED	String	20	N	使用 LMS 積分 For LOYALTY_TYPE=LM only
LMS_BATCH_NO	String	10	N	For LOYALTY_TYPE=LM only
LMS_APPRV	String	10	N	For LOYALTY_TYPE=LM only

5.5 Refund Request

Field	Java Object	Max Length	Required	Description
EVENT_NAME	String	6	Y	REFUND



TXN_ID	String	32	Y	Unique Txn ID generated by ECR
TXN_AMT	Decimal	12	Y	Transaction Amount
PAYMENT_APP_ID	String	10	N	Refer to Payment Application List If empty, Teller will prompt dialog to choose paymentApp. For CC,Amex & QRC only
QRC_VALUE	String	150	N	For CC, it is the Customer Presented QR Code (CPQRC) Value For Wechat/Alipay, it is the Transaction QR/Barcode (QR printed in sale slip or barcode in wallet's transaction history)
MERCHANT_REF	String	10	N	ECR needs to send to N5 only If it needs to print this ref on n5 slip
TXN_DATE	String	4	N	For UP & UP-QR only, Original transaction Date in MMDD format
HOST_REF	String	32	N	For UP & UP-QR only, it is RRN

5.6 Refund Response

Field	Java Object	Max Length	Required	Description
EVENT_NAME	String	11	Y	REFUND_RESP
TXN_ID	String	32	Y	Unique Txn ID per each request defined by ECR
STATUS	String	2	Y	Refer to Status Code List
TXN_AMT	Decimal	12	Y	Transaction Amount (Original) However, it may not equal to Authorization Amount
PAYMENT_TYPE	String	10	N	Refer to Payment Type List
RESP_CODE	String	32	N	Only exists if Status = 00/10 For CC, it is the authorization response code (or UPLAN resp code if failed), For Wechat/Alipay, it is the err_code If Status = 00 It is the error code of printing. Z0001 Printer error Z0002 Printer no paper

Below fields will be returned only if STATUS = 00



TRACE_NO	String	32	Y	For CC, it is the trace no on sale slip (e.g. invoice no from TAS) For WECHAT/ALIPAY, it is the 商戶交易單號
TXN_DATE	String	8	Y	Transaction Date in YYYYMMDD format
TXN_TIME	String	8	Y	Transaction Date in hhmmss format
HOST_REF	String	32	Y	For CC, it is RRN For WECHAT/ALIPAY, it is the transaction id from 第三方
AUTH_CODE	String	8	N	For CC only, authorization code
LOCAL_CUR	String	3	N	Transaction Currency code e.g. 344
FOREIGN_CUR	String	3	N	Cardholder Currency code e.g. 156
FXRATE	Decimal		N	Exchange Rate
FOREIGN_AMT	Decimal		N	Foreign Transaction Amount
MID	String	32	Y	For CC: MID For WECHAT/ALIPAY: sub_mch_id
TID	String	32	Y	For CC: TID For WECHAT/ALIPAY: terminal_id
PAN	String	19	N	For CC only Masked PAN
EXPIRY_DATE	String	4	N	For CC only Masked Expiry Date YYMM
Below fields will be returned only if STATUS = 00 && ECR needs to print receipt				
AID	String	32	N	For CC & Amex only
ENTRY_MODE	String	5	N	For CC & Amex only Chip: (I) RF: (C) SWIPE: (S) FALLBACK: (F) QR: (Q) MANUAL: (M)
BATCH_NO	String	6	N	For CC & Amex only
TC	String	30	N	For CC & Amex only
APP	String	30	N	For CC & Amex only



ACQUIRER	String	30	N	For CC & Amex only
CARD_HOLDER	String	100	N	For CC only, when APP = VISA/MasterCard/JCB and when entry mode is not contactless

5.7 Adjust Request

Field	Java Object	Max Length	Required	Description
EVENT_NAME	String	6	Y	ADJUST
TXN_ID	String	32	Y	Original Transaction ID
TOTAL_TXN_AMT	Decimal	12	N	Adjusted Total Txn Amt for NON-DCC (Should provide for either ONE)
TOTAL_FOREIGN_AMT	Decimal	12	N	Adjusted Total Txn Amt for DCC (Should provide for either ONE)
PAYMENT_APP_ID	String	10	N	Refer to Payment Application List(Optional) If empty, Teller will prompt dialog to choose paymentApp. For CC & Amex only
MERCHANT_REF	String	10	N	ECR needs to send to N5 only If it needs to print this ref on n5 slip

5.8 Adjust Response

Field	Java Object	Max Length	Required	Description
EVENT_NAME	String	11	Y	ADJUST_RESP
TXN_ID	String	32	Y	Original Transaction ID
STATUS	String	2	Y	Refer to Status Code List
TXN_AMT	Decimal	12	Y	Transaction Amount (Original) However, it may not equal to Authorization Amount
TIPS	Decimal	12	Y	Finalized Tips
PAYMENT_TYPE	String	10	N	Refer to Payment Type List
RESP_CODE	String	32	N	Only exists if Status = 00/10 For CC, it is the authorization response code
Below fields will be returned only if STATUS = 00				
LOYALTY_TYPE	String	2	N	If used Coupon/loyalty, this field will be returned UP - Uplan



				MT - Merchant Coupon BK - Bank Coupon LM - LMS
CAMPAIGN_ID	String	30	N	If used Coupon/loyalty, the related campaign id will be returned
TRACE_NO	String	32	Y	For CC, it is the trace no on sale slip (e.g. invoice no from TAS) For WECHAT/ALIPAY, it is the 商戶交易單號
TXN_DATE	String	8	Y	Transaction Date in YYYYMMDD format
TXN_TIME	String	8	Y	Transaction Date in hhmmss format
HOST_REF	String	32	Y	For CC, it is RRN
COUPON_ID	String	19	N	Applied if Loyalty Type is UP/MT/BK Coupon ID
AUTH_CODE	String	8	N	For CC only, authorization code
AUTH_AMT	Decimal		N	Txn_amt + tips - discount (refer below)
DISCOUNT	Decimal		N	Applied if Loyalty Type has value Discount Amount
LOCAL_CUR	String	3	N	Transaction Currency code e.g. 344
FOREIGN_CUR	String	3	N	Cardholder Currency code e.g. 156
FXRATE	Decimal		N	Exchange Rate
FOREIGN_AMT	Decimal		N	Foreign Transaction Amount
TENDER	Integer		N	Installment Tender (apply for installment only)
MID	String	32	Y	For CC: MID For WECHAT/ALIPAY: sub_mch_id
TID	String	32	Y	For CC: TID For WECHAT/ALIPAY: terminal_id
PAN	String	19	N	For CC only Masked PAN
EXPIRY_DATE	String	4	N	For CC only Masked Expiry Date YYMM
LOYALTY_REF	String	32	N	MT/BK -商戶訂單號 LM – LMS Reference (12 digits)
Below fields will be returned only if STATUS = 00 && ECR needs to print receipt				
AID	String	32	N	For CC & Amex only
BATCH_NO	String	6	N	For CC & Amex only



TC	String	30	N	For CC & Amex only
APP	String	30	N	For CC & Amex only
ACQUIRER	String	30	N	For CC & Amex only
IF_SIGN	String	1	N	For CC & Amex only
CARD_TAG	String	30	N	For CC, CUP cards only Credit: (C) Debut: (D)
CARD_TYPE	String	5	N	For CC & Amex only Chip: (C) RF: (T) SWIPE: (S) FALLBACK: (F) QR: (Q) MANUAL: (M)
ENTRY_MODE	String	5	N	For CC & Amex only Chip: (I) RF: (C) SWIPE: (S) FALLBACK: (F) QR: (Q) MANUAL: (M)
SIGN_DESC	String	50	N	For CC & Amex only If need customer sign, return "持卡人簽名 CARDHOLDER SIGNATURE". If no need customer confirm, return "無須簽署 NO SIGNATURE REQUIRED".
TERMS	String	300	N	For CC & Amex only PLEASE USE 5.29 PARAMS RQUEST to get below params. #tc1:"本人確認以上交易，同意將其記入本 卡賬戶 I ACKNOWLEDGE SATISFACTORY RECEIPT OF RELATIVE GOODS/SERVICES. I AGREE TO PAY THE ABOVE TOTAL AMOUNT



				(TOGETHER WITH ANY OTHER CHARGES DUE THEREON) ACCORDING TO THE CARD ISSUER AGREEMENT."
DCC_VISA_TERMS	String	300	N	For DCC only, CUP does not support DCC PLEASE USE 5.29 PARAMS REQUEST to get below params. #dcc1:"匯率 Rate : " #dcc2:"Mark[X] Transaction Currency" #dcc3:"THIS RECEIPT COMPLIES WITH VISA RULES AND THE SERVICE IS PROVIDED BY SERVICE PROVIDER I'VE BEEN OFFERED CHOICE OF CURRENCIES INCLUDING" #dcc4:"PERCENT OVER WHOLE SALE RATE." VISA DCC "#dcc1 HKD/USD 0.133500 #dcc2 []HKD Amount USD Amount[] HKD1500.00 USD200.25 #dcc3 HKD*INCL.THREE PT. FIVE ZERO #dcc4" MASTER/JCB DCC "#dcc1 HKD/USD 0.133500 #dcc2 []HKD Amount USD Amount[] HKD1500.00 USD200.25
CARD_HOLDER	String	100	N	For CC only, when APP = VISA/MasterCard/JCB and when entry mode is not contactless

5.9 Preauth Request

Field	Java Object	Max Length	Required	Description
EVENT_NAME	String	7	Y	PRAUTH
TXN_ID	String	32	Y	Unique Txn ID generated by ECR
TXN_AMT	Decimal	12	Y	Transaction Amount
PAYMENT_APP_ID	String	10	Y	Refer to Payment Application List If empty, Teller will prompt dialog to choose paymentApp. For CC & Amex only

MERCHANT_REF	String	10	N	ECR needs to send to N5 only If it needs to print this ref on n5 slip
PAN	String	19	N	For CC/Amex, used for manual input entrymode
EXPIRY_DATE	String	4	N	For CC/Amex, used for manual input entrymode YYMM
CARD_TYPE	String	3	N	VC: Visa Card MC: MasterCard JCB: Japan Credit Bureau Card UP: Union Pay

5.10 Preauth Response

Field	Java Object	Max Length	Required	Description
EVENT_NAME	String	12	Y	PRAUTH_RESP
TXN_ID	String	32	Y	Unique Txn ID per each request defined by ECR
STATUS	String	2	Y	Refer to Status Code List
TXN_AMT	Decimal	12	Y	Transaction Amount
PAYMENT_TYPE	String	10	N	Refer to Payment Type List
RESP_CODE	String	32	N	Only exists if Status = 00/10 For CC, it is the authorization response code (or UPLAN resp code if failed),
Below fields will be returned only if STATUS = 00				
TRACE_NO	String	32	Y	For CC, it is the trace no on sale slip (e.g. invoice no from TAS)
TXN_DATE	String	8	Y	Transaction Date in YYYYMMDD format
TXN_TIME	String	8	Y	Transaction Date in hhmmss format
HOST_REF	String	32	Y	For CC, it is RRN
AUTH_CODE	String	8	N	For CC only, authorization code
MID	String	32	Y	For CC: MID
TID	String	32	Y	For CC: TID
PAN	String	19	N	For CC only Masked PAN
EXPIRY_DATE	String	4	N	For CC only Masked Expiry Date YYMM
Below fields will be returned only if STATUS = 00 && ECR needs to print receipt				
AID	String	32	N	For CC & Amex only



ENTRY_MODE	String	1	N	For CC & Amex only Chip: (I) RF: (C) SWIPE: (S) FALLBACK: (F) QR: (Q) MANUAL: (M)
BATCH_NO	String	6	N	For CC & Amex only
TC	String	30	N	For CC & Amex only
APP	String	30	N	For CC & Amex only
ACQUIRER	String	30	N	For CC & Amex only
IF_SIGN	String	1	N	For CC & Amex only
CARD_TAG	String	30	N	For CC, CUP cards only Credit: (C) Debit: (D)
CARD_TYPE	String	5	N	For CC & Amex only Chip: (C) RF: (T) SWIPE: (S) FALLBACK: (F) QR: (Q) MANUAL: (M)
SIGN_DESC	String	50	N	For CC & Amex only If need customer sign, return "持卡人簽名 CARDHOLDER SIGNATURE". If no need customer confirm, return "無須簽署 NO SIGNATURE REQUIRED".
TERMS	String	300	N	For CC & Amex only PLEASE USE 5.29 PARAMS REQUEST to get below params.



				#tc1:"本人確認以上交易，同意將其記入本卡賬戶 I ACKNOWLEDGE SATISFACTORY RECEIPT OF RELATIVE GOODS/SERVICES. I AGREE TO PAY THE ABOVE TOTAL AMOUNT (TOGETHER WITH ANY OTHER CHARGES DUE THEREON) ACCORDING TO THE CARD ISSUER AGREEMENT."
CARD_HOLDER	String	100	N	For CC only, when APP = VISA/MasterCard/JCB and when entry mode is not contactless

5.11 Offline Request

Field	Java Object	Max Length	Required	Description
EVENT_NAME	String		Y	OFFLINE
TXN_ID	String	32	Y	Unique Txn ID generated by ECR
TXN_AMT	Decimal		Y	Transaction Amount
PAYMENT_APP_ID	String	10	Y	Refer to Payment Application List If empty, Teller will prompt dialog to choose paymentApp. For CC & Amex only
RELATED_ORIG_TXN	Boolean		N	If value = true, this txn will not present read card page. If value = false or empty, this txn will present read card page.
ORIG_TXN_DATE	String	8	N	Original Transaction Date in YYYYMMDD-format [For UP only] If not present, input in Terminal
AUTH_CODE	String	8	N	For CC & Amex only, authorization code
MERCHANT_REF	String	10	N	ECR needs to send to N5 only If it needs to print this ref on n5 slip
PAN	String	19	N	For CC/Amex, used for manual input entrymode
EXPIRY_DATE	String	4	N	For CC/Amex, used for manual input entrymode YYMM



CARD_TYPE	String	3	N	VC: Visa Card MC: MasterCard JCB: Japan Credit Bureau Card UP: Union Pay
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5.12 Offline Response

Field	Java Object	Max Length	Required	Description
EVENT_NAME	String		Y	OFFLINE_RESP
TXN_ID	String	32	Y	Unique Txn ID generated by ECR
STATUS	String	2	Y	Refer to Status Code List
TXN_AMT	Decimal		Y	Transaction Amount
PAYMENT_TYPE	String	10	N	Refer to Payment Type List
RESP_CODE	String	32	N	Only exists if Status = 00/10 For CC, it is the authorization response code
Below fields will be returned only if STATUS = 00				
TRACE_NO	String	32	Y	For CC, it is the trace no on sale slip (e.g. invoice no from TAS)
TXN_DATE	String	8	Y	Transaction Date in YYYYMMDD format
TXN_TIME	String	8	Y	Transaction Date in hhmmss format
HOST_REF	String	32	Y	For CC, it is RRN
AUTH_CODE	String	8	N	For CC only, authorization code
LOCAL_CUR	String	3	N	Transaction Currency code e.g. 344
FOREIGN_CUR	String	3	N	Cardholder Currency code e.g. 156
FXRATE	Decimal		N	Exchange Rate
FOREIGN_AMT	Decimal		N	Foreign Transaction Amount
MID	String	32	Y	For CC: MID For WECHAT/ALIPAY: sub_mch_id
TID	String	32	Y	For CC: TID For WECHAT/ALIPAY: terminal_id
PAN	String	19	N	For CC only Masked PAN
EXPIRY_DATE	String	4	N	For CC only Masked Expiry Date YYMM
Below fields will be returned only if STATUS = 00 && ECR needs to print receipt				
AID	String	32	N	For CC & Amex only
ENTRY_MODE	String	1	N	For CC & Amex only Chip: (I) RF: (C)



				SWIPE: (S) FALLBACK: (F) QR: (Q) MANUAL: (M)
BATCH_NO	String	6	N	For CC & Amex only
TC	String	30	N	For CC & Amex only
APP	String	30	N	For CC & Amex only
IF_SIGN	String	1	N	For CC & Amex only
ACQUIRER	String	30	N	For CC & Amex only
CARD_TAG	String	30	N	For CC, CUP cards only Credit: (C) Debut: (D)
CARD_TYPE	String	5	N	For CC & Amex only Chip: (C) RF: (T) SWIPE: (S) FALLBACK: (F) QR: (Q) MANUAL: (M)
SIGN_DESC	String	50	N	For CC & Amex only If need customer sign, return "持卡人簽名 CARDHOLDER SIGNATURE". If no need customer confirm, return "無須簽署 NO SIGNATURE REQUIRED".
TERMS	String	300	N	For CC & Amex only PLEASE USE 5.29 PARAMS RQUEST to get below params. #tc1:"本人確認以上交易，同意將其記入本 卡賬戶 I ACKNOWLEDGE SATISFACTORY RECEIPT OF RELATIVE GOODS/SERVICES. I AGREE TO PAY THE ABOVE TOTAL AMOUNT



				(TOGETHER WITH ANY OTHER CHARGES DUE THEREON) ACCORDING TO THE CARD ISSUER AGREEMENT."
DCC_VISA_TERMS	String	300	N	For DCC VISA only If use VISA to performs DCC sale, this field will return.
CARD_HOLDER	String	100	N	For CC only, when APP = VISA/MasterCard/JCB and when entry mode is not contactless

5.13 Installment Request

Field	Java Object	Max Length	Required	Description
EVENT_NAME	String		Y	INSTL
TXN_ID	String	32	Y	Unique Txn ID generated by ECR
TXN_AMT	Decimal		Y	Transaction Amount
TENDER	Integer		N	Installment Tender/Period
PAYMENT_APP_ID	String	10	Y	Refer to Payment Application List If empty, Teller will prompt dialog to choose paymentApp. For CC only
QRC_VALUE	String		N	For CC, it is the Customer Presented QR Code (CPQRC) Value
MERCHANT_REF	String	10	N	ECR needs to send to N5 only If it needs to print this ref on n5 slip

5.14 Installment Response

Field	Java Object	Max Length	Required	Description
EVENT_NAME	String		Y	INSTL_RESP
TXN_ID	String	32	Y	Unique Txn ID generated by ECR
STATUS	String	2	Y	Refer to Status Code List
TXN_AMT	Decimal		Y	Transaction Amount
PAYMENT_TYPE	String	10	N	Refer to Payment Application List
RESP_CODE	String	32	N	Only exists if Status = 00/10 For CC, it is the authorization response code
Below fields will be returned only if STATUS = 00				
TRACE_NO	String	32	Y	For CC, it is the trace no on sale slip (e.g. invoice no from TAS)
TXN_DATE	String	8	Y	Transaction Date in YYYYMMDD format



TXN_TIME	String	8	Y	Transaction Date in hhmmss format
HOST_REF	String	32	Y	For CC, it is RRN
AUTH_CODE	String	8	N	For CC only, authorization code
TENDER	Integer		Y	Installment Tender
MID	String	32	Y	For CC: MID For WECHAT/ALIPAY: sub_mch_id
TID	String	32	Y	For CC: TID For WECHAT/ALIPAY: terminal_id
PAN	String	19	N	For CC only Masked PAN
EXPIRY_DATE	String	4	N	For CC only Masked Expiry Date YYMM
AID	String	32	N	For CC & Amex only
ENTRY_MODE	String	5	N	For CC & Amex only Chip: (I) RF: (C) SWIPE: (S) FALLBACK: (F) QR: (Q) MANUAL: (M)
BATCH_NO	String	6	N	For CC & Amex only
TC	String	30	N	For CC & Amex only
APP	String	30	N	For CC & Amex only
ACQUIRER	String	30	N	For CC & Amex only
IF_SIGN	String	1	N	For CC & Amex only
CARD_TAG	String	30	N	For CC, CUP cards only Credit: (C) Debit: (D)



CARD_TYPE	String	5	N	For CC & Amex only Chip: (C) RF: (T) SWIPE: (S) FALLBACK: (F) QR: (Q) MANUAL: (M)
SIGN_DESC	String	50	N	For CC & Amex only If need customer sign, return "持卡人簽名 CARDHOLDER SIGNATURE". If no need customer confirm, return "無須簽署 NO SIGNATURE REQUIRED".
TERMS	String	300	N	For CC & Amex only
CARD_HOLDER	String	100	N	For CC only, when APP = VISA/MasterCard/JCB and when entry mode is not contactless

5.15 Settlement Request

Field	Java Object	Max Length	Required	Description
EVENT_NAME	String		Y	SETTLE
MERCHANT_REF	String	10	N	ECR needs to send to N5 only If it needs to print this ref on n5 slip

5.16 Settlement Response

Field	Java Object	Max Length	Required	Description
EVENT_NAME	String		Y	SETTLE_RESP
STATUS	String	2	Y	Refer to Status Code List

5.17 Scan Request

Field	Java Object	Max Length	Required	Description
EVENT_NAME	String		Y	SCAN

5.18 Scan Response

Field	Java Object	Max Length	Required	Description
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EVENT_NAME	String		Y	SCAN_RESP
STATUS	String	2	Y	Refer to Status Code List
CODE_VALUE	String		N	Return if status = 00 The bar code/qr code value scanned by terminal

5.19 Last Transaction Retrieval

Field	Java Object	Max Length	Required	Description
EVENT_NAME	String		Y	RETRIEVAL
TXN_ID	String	32	Y	Unique Txn ID per each request defined by ECR

5.20 Last Transaction Retrieval Response

Field	Java Object	Max Length	Required	Description
EVENT_NAME	String		Y	RETRIEVAL_RESP
TXN_TYPE	String		Y	SALE/REFUND/PREAUTH/OFFLINE
TXN_ID	String	32	Y	Unique Txn ID per each request defined by ECR
STATUS	String	2	Y	Refer to Status Code List
Below fields will be returned only if STATUS = 00				
TXN_STATUS	String	2	Y	Refer to Transaction Status Code List
TXN_AMT	Decimal		Y	Transaction Amount (Original) However, it may not equal to Authorization Amount
TIPS	Decimal		N	Applied if tip is enabled
PAYMENT_TYPE	String	10	N	Refer to Payment Type List
RESP_CODE	String	32	N	Only exists if Status = 00/10 It is the authorization response code (or UPLAN/LMS resp code if failed), For WECHAT/ALIPAY, it is the err_code
LOYALTY_TYPE	String	2	N	If used Coupon/loyalty, this field will be returned UP - Uplan MT - Merchant Coupon BK - Bank Coupon LM - LMS
CAMPAIGN_ID	String	30	N	If used Coupon/loyalty, the related campaign id will be returned



TRACE_NO	String	32	Y	For CC, it is the trace no on sale slip (e.g. invoice no from TAS) For WECHAT/ALIPAY, it is the 商戶交易單號
TXN_DATE	String	8	Y	Transaction Date in YYYYMMDD format
TXN_TIME	String	8	Y	Transaction Date in hhmmss format
HOST_REF	String	32	Y	For CC, it is RRN For WECHAT/ALIPAY, it is the transaction id from 第三方
COUPON_ID	String	19	N	Applied if Loyalty Type is UP/MT/BK Coupon ID
AUTH_CODE	String	8	N	For CC only, authorization code
TENDER	Integer		N	Installment Tender (apply for installment only)
AUTH_AMT	Decimal		N	Txn_amt + tips - discount (refer below)
DISCOUNT	Decimal		N	Applied if Loyalty Type has value Discount Amount
LOCAL_CUR	String	3	N	Transaction Currency code e.g. 344
FOREIGN_CUR	String	3	N	Cardholder Currency code e.g. 156
FXRATE	Decimal		N	Exchange Rate
FOREIGN_AMT	Decimal		N	Foreign Transaction Amount of AUTH_AMT
MID	String	32	Y	For CC: MID For WECHAT/ALIPAY: sub_mch_id
TID	String	32	Y	For CC: TID For WECHAT/ALIPAY: terminal_id
PAN	String	19	N	For CC only Masked PAN
EXPIRY_DATE	String	4	N	For CC only Masked Expiry Date YYMM
LOYALTY_REF	String	32	N	MT/BK -商戶訂單號 LM – LMS Reference (12 digits)
Below fields will be returned only if STATUS = 00 && ECR needs to print receipt				
AID	String	32	N	For CC & Amex only
BATCH_NO	String	6	N	For CC & Amex only
TC	String	30	N	For CC & Amex only
ACQUIRER	String	30	N	For CC & Amex only



IF_SIGN	String	1	N	For CC & Amex only
CARD_TAG	String	30	N	For CC, CUP cards only Credit: (C) Debut: (D)
CARD_TYPE	String	5	N	For CC & Amex only Chip: (C) RF: (T) SWIPE: (S) FALLBACK: (F) QR: (Q) MANUAL: (M)
ENTRY_MODE	String	5	N	For CC & Amex only Chip: (I) RF: (C) SWIPE: (S) FALLBACK: (F) QR: (Q) MANUAL: (M)
SIGN_DESC	String	50	N	For CC & Amex only If need customer sign, return "持卡人簽名 CARDHOLDER SIGNATURE". If no need customer confirm, return "無須簽署 NO SIGNATURE REQUIRED".
TERMS	String	300	N	For CC & Amex only
TVR	String	10	N	For CC & Amex only
TSI	String	4	N	For CC & Amex only
DCC_VISA_TERMS	String	300	N	For DCC only, CUP doest not support DCC PLEASE USE 5.29 PARAMS RQUEST to get below params. #dcc1:"匯率 Rate : " #dcc2:"Mark[X] Transaction Currency"



				#dcc3: "THIS RECEIPT COMPLIES WITH VISA RULES AND THE SERVICE IS PROVIDED BY SERVICE PROVIDER I'VE BEEN OFFERED CHOICE OF CURRENCIES INCLUDING" #dcc4: "PERCENT OVER WHOLE SALE RATE." VISA DCC "#dcc1 HKD/USD 0.133500 #dcc2 [] HKD Amount USD Amount [] HKD1500.00 USD200.25 #dcc3 HKD*INCL.THREE PT. FIVE ZERO #dcc4" MASTER/JCB DCC "#dcc1 HKD/USD 0.133500 #dcc2 [] HKD Amount USD Amount [] HKD1500.00 USD200.25
DCC_MID	String	32	N	DCC only Displayed under MID 
REMAINED_VALUE	String	20	N	For OPS only



SERIAL_NO	String	10	N	For OPS only
LMS_ORIGINAL_BAL	String	20	N	原有 LMS 積分 For LOYALTY_TYPE=LM only
LMS_BAL	String	20	N	結餘 LMS 積分 For LOYALTY_TYPE=LM only
LMS_USED	String	20	N	使用 LMS 積分 For LOYALTY_TYPE=LM only
LMS_BATCH_NO	String	10	N	For LOYALTY_TYPE=LM only
LMS_APPRV	String	10	N	For LOYALTY_TYPE=LM only

5.21 Link Test

Field	Java Object	Max Length	Required	Description
EVENT_NAME	String		Y	ECHO

5.22 Link Test Response

Field	Java Object	Max Length	Required	Description
EVENT_NAME	String		Y	ECHO_RESP
STATUS	String	2	Y	Refer to Status Code List

5.23 Print Request

Field	Java Object	Max Length	Required	Description
EVENT_NAME	String		Y	PRINT
TXN_ID	String	32	N	If empty, N5 will find last successful transaction and print receipt If not empty, put original transaction ID to print receipt
IS_REPRINT	Boolean		Y	
MERCHANT_REF	String	10	N	ECR needs to send to N5 only If it needs to print this ref on n5 slip



5.24 Print Response

Field	Java Object	Max Length	Required	Description
EVENT_NAME	String		Y	PRINT_RESP
STATUS	String	2	Y	Refer to Status Code List

5.25 Abort Request

Field	Java Object	Max Length	Required	Description
EVENT_NAME	String		Y	ABORT

5.26 Abort Response

Field	Java Object	Max Length	Required	Description
EVENT_NAME	String		Y	ABORT_RESP
STATUS	String	2	Y	Refer to Status Code List

5.27 Params Request

Field	Java Object	Max Length	Required	Description
EVENT_NAME	String		Y	PARAM

5.28 Params Response

Field	Java Object	Max Length	Required	Description
EVENT_NAME	String		Y	PARAM_RESP
STATUS	String	2	Y	Refer to Status Code List
Below fields will be returned only if STATUS = 00				
PARAMS	String		Y	Json String <pre>{ "tc1": "xxxxx", "dcc1": "xxxxx", "dcc2": "xxxxx", "dcc3": "xxxxx", "dcc4": "xxxxx" }</pre>

5.29 Batch Enquiry Request

Field	Java Object	Max Length	Required	Description
EVENT_NAME	String		Y	BATCH_ENQUIRY
SETTLE_DATE	String	8	Y	Transaction Date in YYYYMMDD format

5.30 Batch Enquiry Response

Field	Java Object	Max Length	Required	Description
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EVENT_NAME	String		Y	BATCH_ENQUIRY_RESP
STATUS	String	2	Y	Refer to Status Code List
Below fields will be returned only if STATUS = 00				
RESULT	String		Y	Sample: <pre>{ "EVENT_NAME":"BATCH_ENQUIRY_RESP", "RESULT":["000001","000002"], "STATUS":"00" }</pre>

5.31 Settle Enquiry Request

Field	Java Object	Max Length	Required	Description
EVENT_NAME	String		Y	SETTLE_ENQUIRY
BATCH_ID	String	6	Y	

5.32 Settle Enquiry Response

Field	Java Object	Max Length	Required	Description
EVENT_NAME	String		Y	SETTLE_ENQUIRY_RESP
STATUS	String	2	Y	Refer to Status Code List
Below fields will be returned only if STATUS = 00				
RESULT	String		Y	Sample: <pre>{ "EVENT_NAME": "SETTLE_ENQUIRY_RESP", "STATUS": "00", "RESULT": [{ "MID": "011603999000001", "RC": "0", "OC": "0", "TYPE": "UP", "VRC": "0", "RA": "0.00", "SC": "1", "OA": "0.00", "SA": "0.05", "VSC": "0", "VOC": "0", "VSA": "0.00", "TID": "21700511", "VRA": "0.00", </pre>



```
"VOA": "0.00"
}, {
  "MID": "010601994000005",
  "RC": "0",
  "OC": "0",
  "TYPE": "DCC-VC",
  "VRC": "0",
  "RA": "0.00",
  "SC": "1",
  "OA": "0.00",
  "SA": "5.00",
  "VSC": "0",
  "VOC": "0",
  "VSA": "0.00",
  "TID": "81720400",
  "VRA": "0.00",
  "VOA": "0.00"
}, {
  "MID": "010601994000005",
  "RC": "0",
  "OC": "0",
  "TYPE": "DCC-MC",
  "VRC": "0",
  "RA": "0.00",
  "SC": "0",
  "OA": "0.00",
  "SA": "0.00",
  "VSC": "0",
  "VOC": "0",
  "VSA": "0.00",
  "TID": "81720400",
  "VRA": "0.00",
  "VOA": "0.00"
}]
}
```

5.33 Summary Enquiry Request

Field	Java Object	Max Length	Required	Description
EVENT_NAME	String		Y	SUMMARY_ENQUIRY
TXN_DATE_START_TO_END	String	8	N	If empty, will return all unsettled txns. otherwise will only return txns in specific period.



				TXN_DATE_START_TO_END is in YYYYMMDD-YYYYMMDD format e.g. 20200602-20200702
--	--	--	--	---

5.34 Summary Enquiry Response

Field	Java Object	Max Length	Required	Description
EVENT_NAME	String		Y	SUMMARY_ENQUIRY_RESP
STATUS	String	2	Y	Refer to Status Code List
Below fields will be returned only if STATUS = 00				
RESULT	String		Y	Sample: <pre>{ "EVENT_NAME": "UNSETTLED_ENQUIRY_RESP", "STATUS": "00", "RESULT": [{ "RC": "0", "OC": "0", "TYPE": "UP", "VRC": "0", "RA": "0.00", "SC": "1", "OA": "0.00", "SA": "0.05", "VSC": "0", "VOC": "0", "VSA": "0.00", "TID": "21700511", "VRA": "0.00", "VOA": "0.00" }, { "RC": "0", "OC": "0", "TYPE": "VC", "VRC": "0", "RA": "0.00", "SC": "1", "OA": "0.00", "SA": "5.00", "VSC": "0", "VOC": "0", "VSA": "0.00", "TID": "81720400", "VRA": "0.00", "VOA": "0.00" } }</pre>



				<pre>}, { "RC": "0", "OC": "0", "TYPE": "MC", "VRC": "0", "RA": "0.00", "SC": "0", "OA": "0.00", "SA": "0.00", "VSC": "0", "VOC": "0", "VSA": "0.00", "TID": "81720400", "VRA": "0.00", "VOA": "0.00" } }</pre>
--	--	--	--	---

5.35 Create Order Request

Field	Java Object	Max Length	Required	Description
EVENT_NAME	String		Y	"CREATE_ORDER"
PAYMENT_APP_ID	String	10	Y	QRC only. This function is used for merchant QR. User can use their own wallet to scan QR and perform payment.
TXN_ID	String	32	Y	Unique Txn ID generated by ECR
TXN_AMT	Decimal		Y	Transaction Amount
PAYMENT_TYPE	String	10	Y	Support Wechat Pay, Alipay, UnionPay QR

5.36 Create Order Response

Field	Java Object	Max Length	Required	Description
EVENT_NAME	String		Y	"CREATE_ORDER_RESP"
TXN_ID	String	32	Y	Unique Txn ID generated by ECR
STATUS	String	2	Y	00: SUCCESS 10: FAIL
TXN_AMT	Decimal		Y	Payment Amount



PAYMENT_TYPE	String	10	N	Support Wechat Pay, Alipay, UnionPay QR
Below fields will be returned only if STATUS = 00				
MID	String	32	Y	For CC: MID For Wechat/Alipay: sub_mch_id
TID	String	32	Y	For CC: TID For Wechat/Alipay: terminal_id
RESP_CODE	String	32	N	Only exists if Status = 00/10
SERIAL_NO	String	10	Y	POS Serial No. of the Device
QRC_VALUE	String	256	Y	QR code

5.37 Health Check Request

Field	Java Object	Max Length	Required	Description
EVENT_NAME	String		Y	"HEALTH_CHECK"

5.38 Health Check Response

Field	Java Object	Max Length	Required	Description
EVENT_NAME	String		Y	"HEALTH_CHECK_RESP"
STATUS	String	2	Y	00: SUCCESS 10: FAIL
Below fields will be returned only if STATUS = 00				
RESULT	String	256	Y	<pre>[{ "name": "TMS Service connection detection", "url": "https://www.etpspay.com/TMSWEB/GetTMSParams.do", "result": "success" }, { "name": "TMS Service connection detection", "url": "https://www.etpspay.com/TMSWEB/GetTMSParams.do", "result": "fail" }]</pre>

5.39 Settlement Complete Request

Field	Java Object	Max Length	Required	Description
EVENT_NAME	String		Y	"SETTLEMENT_COMPLETE"

5.40 Settlement Complete Response

No response for settlement complete, POS will restart after receive this request.



6. Appendix

6.1 Payment Application List

Payment Application	Description
CC	Credit Card
QRC	QR/Barcode Payment (e.g. Wechat, Alipay, VMJ QR, UP QR)
CPN	Coupon Usage
AE	AMEX
DC	Diner Club (Not Support Yet)
FPS	Fast Payment Service
OPS	Octopus
EPS	Easy Pay System

6.2 Payment Type List

Payment Type	Description	Support Country
VC	Visa Card	SEA
VC-QR	Visa Card QR	SEA
MC	Master Card	SEA
MC-QR	Master Card QR	SEA
JCB	JCB	SEA
JCB-QR	JCB QR	SEA
UP	Unionpay	SEA
UP-QR	Unionpay QR	SEA



AE	AMEX	SEA
ALP	Alipay	SEA
WCP	Wechat Pay	SEA
FPS	Fast Payment Service.	SEA
OPS	Octopus	SEA
EPS	EPS	SEA
MO-BOCPAY	MACAU BOCPAY	MACAU
TFPAY	Tai Fung Pay(丰付宝)	MACAU
UEPAY	UEPAY(极易付)	MACAU
ICBCEPAY	工行 e 支付	MACAU
CGBPAY	广发银行	MACAU
LUSOPAY	国际银行	MACAU
MPAY	澳门通	MACAU

6.3 Status Code Table

	SALE_RESP	VOID_RESP	SETTLE_RESP	REFUND_RESP	PREAUTH_RESP
00: Approved/Accepted by host	✓	✓	✓	✓	✓
02: Refund Processing/ Accepted by host Only for QRC Noted: ECR should treat 02 as successful transaction				✓	
03: Confirming/ Processing	✓	✓		✓	
09: Cancelled by user	✓	✓	✓	✓	✓



10: Declined by host	✓	✓	✓ [Partial Settlement]	✓	✓
11: Reversal by User (Applied for Wechat/Alipay Only)	✓				
12: Transaction has already voided/completed/reversed		✓			
13: Transaction does not find		✓			
14: Tender does not support					
90: Communication/Format Error	✓	✓	✓	✓	✓
91: Function or Payment Type does not supported	✓	✓	✓	✓	✓
97: ECR busy	✓	✓	✓	✓	✓
98: Timeout	✓	✓	✓	✓	✓
99: Unknown Error	✓	✓	✓	✓	✓

	OFFLINE_RESP	ADJUST_RESP	INSTL_RESP	SCAN_RESP	RETRIEVAL_RESP
00: Approved/Accepted by host	✓	✓	✓	✓	✓
03: Confirming/Processing					
09: Cancelled by user	✓	✓	✓	✓	
10: Declined by host	✓	✓	✓		
11: Reversal by User (Applied for Wechat/Alipay Only)					
12: Transaction has already voided/completed/reversed	✓				



13: Transaction does not find	✓	✓			✓
14: Tender does not support			✓		
90: Communication/Format Error	✓	✓	✓	✓	✓
91: Function or Payment Type does not supported	✓	✓	✓	✓	✓
97: ECR busy	✓	✓	✓	✓	✓
98: Timeout	✓	✓	✓	✓	✓
99: Unknown Error	✓	✓	✓	✓	✓

	TEST_RESP	REVERSAL_RE SP	ADD_POINT_RE SP	ABORT_RESP	PARAM_RESP
00: Approved/Accepted by host	✓	✓	✓	✓	✓
03: Confirming/Processing					
09: Cancelled by user			✓		
10: Declined by host			✓	✓	✓
11: Reversal by User (Applied for Wechat/Alipay Only)					
12: Transaction has already voided/completed/reversed		✓			
13: Transaction does not find		✓			
14: Tender does not support		✓			
90: Communication/Format Error	✓				
91: Function or Payment Type does not supported			✓		



97: ECR busy			✓		✓
98: Timeout	✓		✓		
99: Unknown Error			✓		

	BATCH_ENQUIRY_RESP	SETTLE_ENQUIRY_RESP	SUMMARY_ENQUIRY_RESP	CREATE_ORDER_RESP
00: Approved/Accepted by host	✓	✓	✓	✓
03: Confirming/Processing				
09: Cancelled by user				
10: Declined by host				✓
11: Reversal by User (Applied for Wechat/Alipay Only)				
12: Transaction has already voided/completed/reversed				
13: Transaction does not find				
14: Tender does not support				
90: Communication/Format Error	✓	✓	✓	✓
91: Function or Payment Type does not supported				
97: ECR busy	✓	✓	✓	✓
98: Timeout				
99: Unknown Error	✓	✓	✓	✓



6.4 Event to Payment Type Support List

	CC	QRC	CPN	AE	FPS	OPS	EPS
SALE	✓	✓	✓	✓	✓	✓	✓
VOID	✓	✓		✓			✓
SETTLE	✓	✓		✓		✓	✓
REFUND	✓	✓		✓			
PREAUTH	✓			✓			
OFFLINE	✓			✓			
ADJUST	✓			✓			
INSTL	✓			✓			
RETRIEVAL	✓	✓		✓	✓	✓	✓
PRINT	✓	✓		✓	✓	✓	✓
ABORT	✓	✓		✓	✓	✓	✓
BATCH_EN QUIRY	✓	✓		✓	✓	✓	✓
SETTLE_EN QUIRY	✓	✓		✓	✓	✓	✓
SUMMARY_ ENQUIRY	✓	✓		✓	✓	✓	✓
CREATE_O RDER		✓					

	MO- BOCPAY	TFPAY	UEPAY	ICBCEPAY	CGBPAY	LUSOPAY	MPAY
SALE	✓	✓	✓	✓	✓	✓	✓
VOID	✓	✓	✓	✓	✓	✓	✓
SETTLE	✓	✓	✓	✓	✓	✓	✓
REFUND	✓	✓	✓	✓	✓	✓	✓

6.5 Transaction Status Code List

Transaction Status	Description
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A	Approved/Accepted by host
P	Confirming/Processing
V	Voided
S	Settled (Only A will change to S after settlement)
F	Declined/Cancelled/Reversed/Operational Timeout
T	Octopus read card timeout

6.6 Settle Type List

Payment Type	Description
VC	Visa Card
MC	Master Card
JCB	JCB
UP	Unionpay
UP-QR	Unionpay QR
AE	AMEX
ALP	Alipay
WCP	Wechat Pay
FPS	Fast Payment Service.
OPS	Octopus
EPS	EPS
DCC-VC	DCC + Visa Card
DCC-MC	DCC + Master Card
LMS-UP	Loyalty + Unionpay
LMS-VC	Loyalty + Visa Card



UPLAN	UPLAN
MO-BOCPAY	MACAU BOCPAY
TFPAY	Tai Fung Pay(丰付宝)
UEPAY	UEPAY(极易付)
ICBCEPAY	工行 e 支付
CGBPAY	广发银行
LUSOPAY	国际银行
MPAY	澳门通